Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **[Your team name]**
b. Team leader name: **[Your full name]**
c. Problem Statement: **Retail & Commerce: Intelligent Customer and Business Experiences. Etalase makes WhatsApp & Instagram sellers discoverable and buyable by AI shopping agents**

Brief about the idea

Etalase turns a WhatsApp seller's product photos and chats into a storefront that AI shopping agents can find, check stock on and buy from, with every order landing back in WhatsApp.

Why now

AI agents already check out through Google's Universal Commerce Protocol (UCP) in the US and Australia, with India in testing. Morgan Stanley expects US$190–385B of US online spending to go through shopping agents by 2030. Platforms are making their own merchants agent-ready; independent chat sellers have no path in.

The problem

Agents only read structured catalogs. Across Southeast Asia, small sellers run on chat apps such as WhatsApp, LINE, Zalo and Messenger, with catalogs in photos and messages, so agents cannot see them. When agent checkout arrives, that demand flows to marketplaces and big brands instead.

Our solution

Gemini builds the catalog from photos, voice notes and spreadsheets. Etalase publishes it over MCP and the UCP checkout API, reserves stock at checkout and takes QRIS payments. The seller confirms and ships from WhatsApp, with no new app to learn.

US$185B

Southeast Asia e-commerce GMV in 2025; consumer interest in AI is 3× the global average

95%

of Indonesian online businesses sell through chat apps such as WhatsApp; only 18% use marketplaces (BPS, 2023)

~1 in 2

Southeast Asian shoppers have bought or sold on social media or messaging apps (2019 survey)

11%

of small and mid-size merchants are ready for AI shopping agents, vs 19% of large merchants (2026)

Sources: e-Conomy SEA 2025 (Google, Temasek, Bain, Nov 2025); BPS Statistik E-Commerce 2023 (pub. Jan 2025); Facebook × BCG conversational commerce study (Nov 2019); PYMNTS × Visa, 2026 Global Digital Shopping Index (Sep 2026); Morgan Stanley, "Here Come the Shopping Bots". Links in the repo: docs/15-evidence.md.

Opportunities

How is it different?

- Agent-readiness tools (Era, ShopAgentic, AgentReady) serve Shopify-style stores that already have product data
- Marketplaces and platforms make sellers agent-ready only inside their own ecosystem, at their fees
- WhatsApp chatbots answer human buyers, not AI agents
- **Etalase serves sellers with no store system and opens them to any agent via open protocols**

How does it solve the problem?

- Builds the catalog from what sellers already have: photos, voice notes, chats, spreadsheets
- Publishes it in agent-readable form: MCP tools, UCP checkout API, schema.org data
- Runs a safe checkout: stock reservation, seller confirmation, QRIS payment, courier rates
- Keeps the seller in WhatsApp, with no new app to learn

USP of the solution

- **From WhatsApp photos to an agent-ready catalog in minutes**
- Brings chat-first micro-sellers into AI commerce with a channel they own, paid only per order
- One core with pluggable country packs for payment, shipping and OMS
- Privacy by design: tenant-isolated AI and minimal data, built with Indonesia's PDP Law in mind

List of features offered by the solution

AI catalog builder

Built

Gemini reads product photos, voice notes, captions and CSV files, then drafts products, variants, prices and weights for the seller to confirm.

Agent-ready storefront

Built

Each shop is exposed through MCP tools (search, shipping, checkout), a UCP profile and checkout API, and schema.org data, with a 0–100 readiness score.

WhatsApp seller agent

Built

An ADK agent lets sellers update stock and prices by chat, accept orders and send tracking numbers. Order buttons never go through the model.

Safe agent checkout

Built · QRIS mocked

Stock is reserved at checkout and low-confidence stock is confirmed with the seller first: 0 units oversold when 100 buyers race.

Shipping & payouts

Mocked in demo

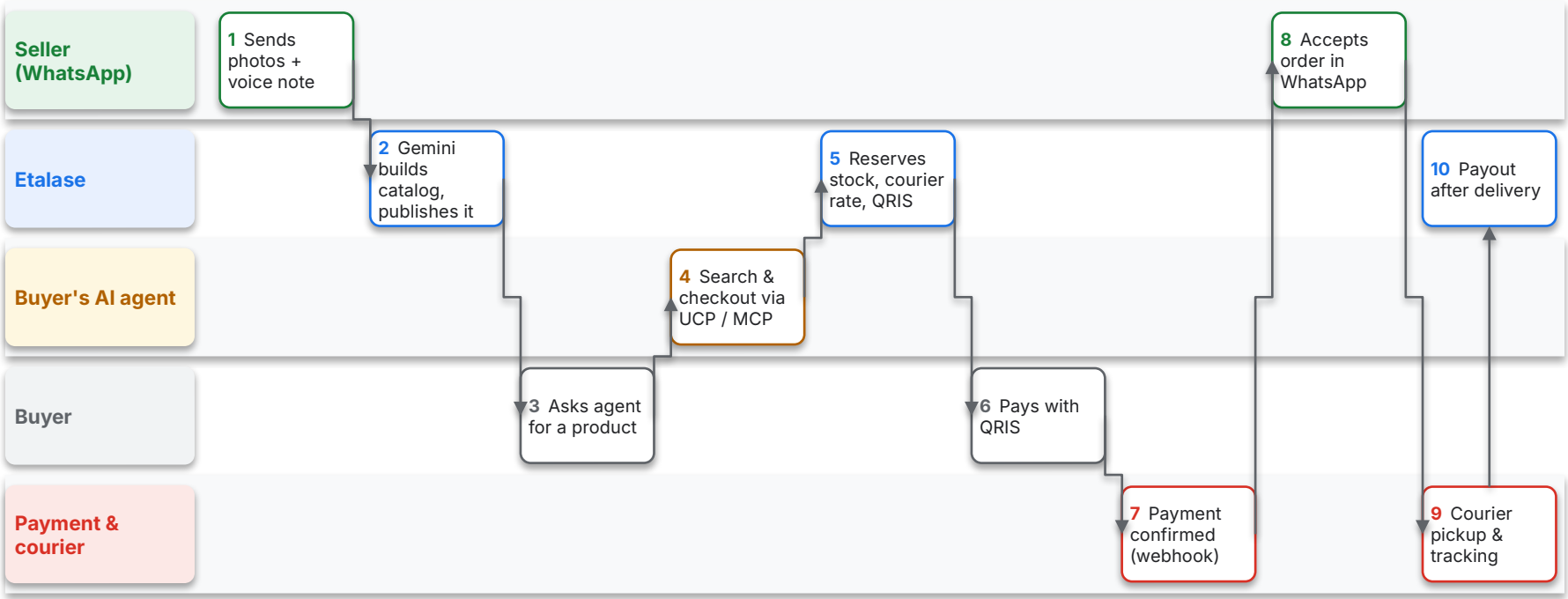
Courier rates, delivery confirmed with the courier, then payout minus commission. Biteship and Xendit adapters are wired; the demo runs on mocks.

Connectors

Built · more next

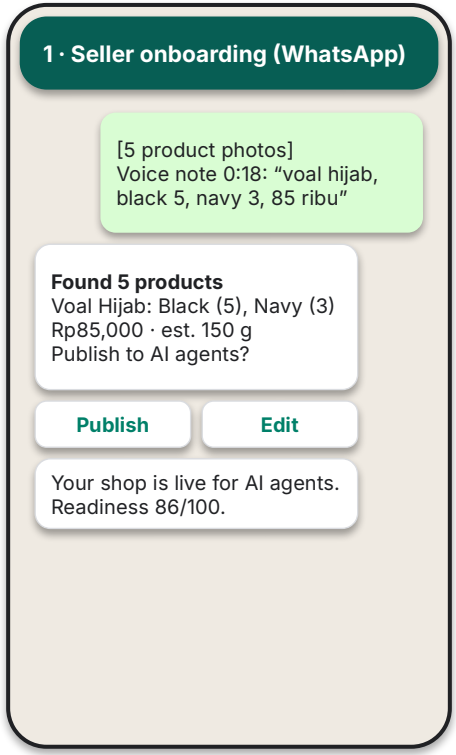
Instagram post import and an OpenAPI Connector API for website builders work today. Google Sheets, OMSs and WooCommerce come next.

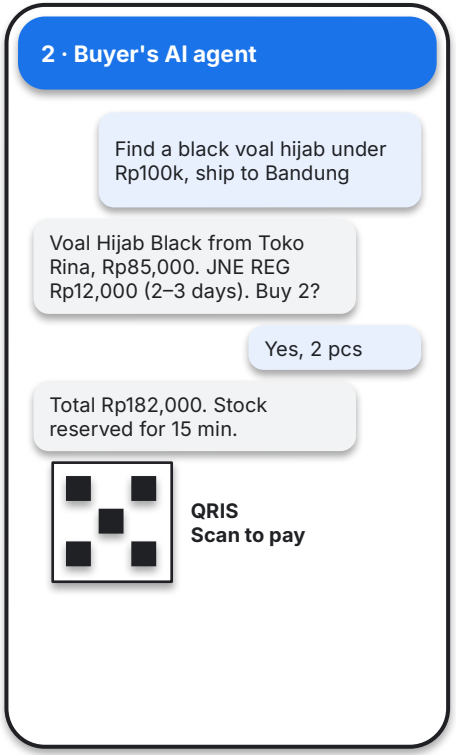
Process flow diagram or Use-case diagram

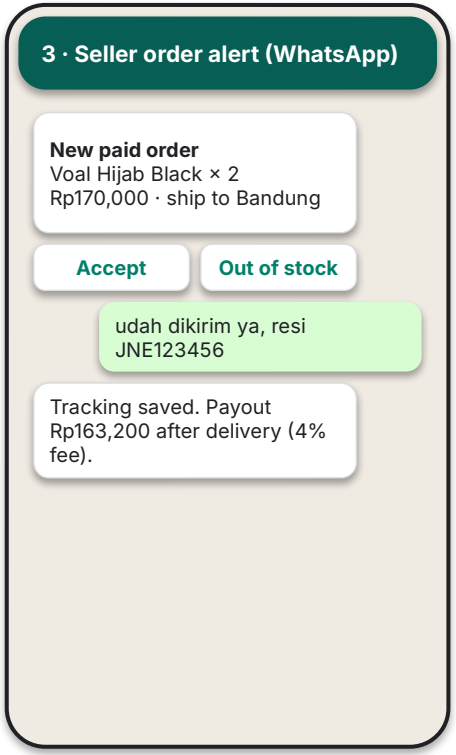


If stock confidence is low (stale or nearly sold out), the seller confirms stock before the QRIS link is sent. Stock reservations expire after 15 minutes if unpaid.

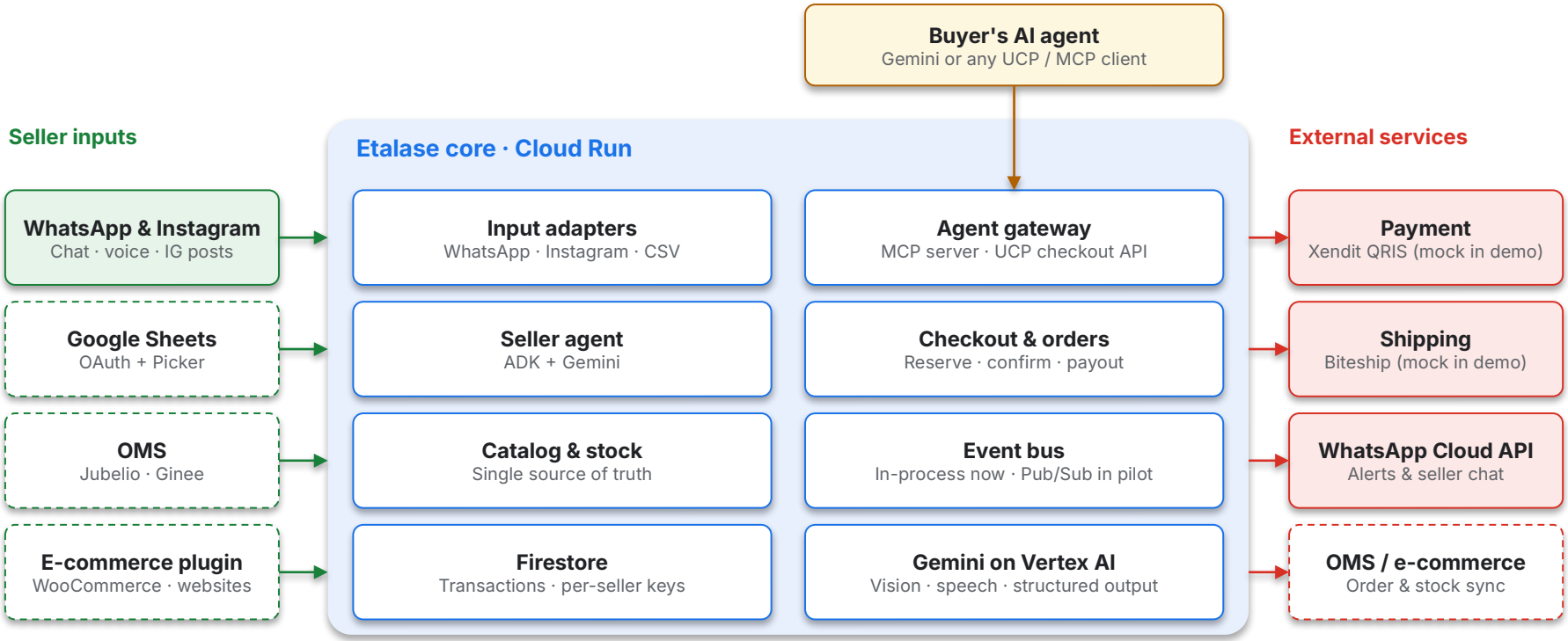
Wireframes/Mock diagrams of the proposed solution (optional)







Architecture diagram of the proposed solution



Solid = hackathon MVP · dashed = roadmap. Country pack = { currency, payment, shipping, tax } adapters: a new market adds one config entry, not a new core. Each seller has one stock owner: Etalase, the OMS or the store.

Technologies to be used in the solution

AI & agents

- Gemini on Vertex AI: photos, voice and text to catalog with structured output
- ADK seller agent with tenant-scoped tools
- ADK buyer agent calling our MCP tools (demo)

Agentic commerce

- MCP server: search, shipping, checkout, status
- UCP profile and checkout API (/ucp/v1)
- schema.org product data per store
- Next: Merchant Center feed, AP2 payments

Google Cloud runtime

- Cloud Run: API, MCP, webhooks, dashboard
- Firestore store with transactions
- Secret Manager for every key
- Pilot: Pub/Sub, Cloud Tasks, Storage, KMS

Integrations

- WhatsApp Business Cloud API
- Xendit QRIS and Biteship adapters
- Instagram API (catalog import)
- OpenAPI Connector API

Seller dashboard

- Lightweight web dashboard on Cloud Run
- WhatsApp one-time-code sign-in
- Readiness, orders and payouts

Security & operations

- Tenant isolation, including AI context
- Signed webhooks, fail-closed startup checks
- Rate limits, OTP lockout, CSP
- Built with Indonesia's PDP Law in mind

Estimated implementation cost (optional)

Pilot assumption: 100 sellers · 3,000 products · 1,000 orders per month. Rough estimates, to be validated with the Google Cloud Pricing Calculator.

Component	Basis	Est. / month
Cloud Run, Firestore, Storage, Pub/Sub	Low pilot traffic, mostly within free tiers	US$10–30
Gemini on Vertex AI	~3,000 catalog extractions + ~2,000 seller-agent turns (buyers' agents run on their own models)	US$15–60
WhatsApp Cloud API	~4 utility messages per order × 1,000 orders at ~IDR 366–460	~US$90–115
Biteship rates API	~5,000 rate checks at ~IDR 5 each	< US$2
QRIS payments	0% MDR for micro sellers ≤ Rp500k (Bank Indonesia); gateway fee paid by buyer	Pass-through
Total platform cost		**≈ US$120–210**

Unit economics per order

≈ IDR 2–3.5k

platform cost per order in the pilot (WhatsApp is most of it)

IDR 6.8k

commission at 4% on an IDR 170k order. Sellers pay only when an order sells.

Price references: Meta WhatsApp Business Platform per-message pricing (utility rate in Indonesia via Qontak, Oct 2026); Biteship Rates API pricing; Bank Indonesia QRIS MDR rules (Kontan, Oct 2026); Vertex AI Gemini 2.5 Flash list prices. FX: US$1 ≈ IDR 16,000. Details: docs/10.

Snapshots of the prototype

Prototype in progress. Screenshots from the deployed build will replace these frames before the final submission.

Screenshot to be added

Seller onboarding in WhatsApp

Screenshot to be added

Buyer agent search & checkout

Screenshot to be added

Order alert & seller dashboard

Prototype Performance report/Benchmarking

214 automated tests pass today (stock races, payments, webhooks, OTP abuse). TBD results come from the Gemini run on Cloud Run.

Metric	How we measure	Target	Result
Catalog extraction accuracy	50 labelled products: title, variant, price, stock	≥ 90% of fields correct	TBD
Catalog build time	Photos + voice note to published product	< 15 s per product	TBD
Agent tool latency	p95 of search_products and create_checkout	< 2 s	TBD
Overselling under load	100 concurrent reservations for the last unit	0 units oversold	Pass (in-memory)
Seller alert speed	Payment webhook to WhatsApp message	< 5 s	TBD
Duplicate & late payments	Same event twice; payment after expiry	No double change; auto refund	Pass
Stock-update understanding	Intent accuracy on 100 real seller messages	≥ 95%	TBD

Additional Details/Future Development (if any)

1 2 3 4

Hackathon MVP

- WhatsApp, Instagram and CSV inputs
- MCP gateway, UCP checkout API
- Demo buyer agent on Gemini (ADK)
- Mock QRIS and courier; Xendit, Biteship wired

Indonesia pilot

- 50–100 sellers onboarded
- Firestore, Pub/Sub, Cloud Tasks
- KYC and payouts via sub-accounts
- Sheets sync, Merchant Center feed

More channels

- OMS adapters: Jubelio, Ginee
- E-commerce plugins, WooCommerce first
- Shopee and TikTok Shop API sync
- Sales analytics for sellers

Regional

- Country packs: MY, PH, TH, VN
- Chat adapters: LINE, Zalo, Messenger
- Stripe in supported markets
- Cross-border shipping and duties

What we will validate next

Seller willingness to pay per order · timing of UCP checkout in Indonesia · payment licensing through partners, so Etalase never holds seller funds · stock accuracy for sellers who do not track stock

Provide links to your:

GitHub public repository	[Add GitHub link: github.com/<team>/etalase]
Demo video (3 minutes)	[Add YouTube / Drive link]
Final product link	[Add Cloud Run URL]

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Thank you

